

Context

```
using MiniProject2.Models;
using Microsoft.EntityFrameworkCore;

namespace MiniProject2.Context
{
    public class AppDbContext : DbContext
    {
        public AppDbContext(DbContextOptions<AppDbContext> options) : base(options)
        {

        }

        public DbSet<CustomerDetails> customerDet { get; set; }
        public DbSet<CustomerAccount> customerAcc { get; set; }
    }
}
```

Account Controller

```
using Microsoft.AspNetCore.Authorization;
using Microsoft.AspNetCore.Mvc;
using MiniProject2.Context;
using MiniProject2.Models;

namespace MiniProject2.Controllers
{
    public class CustomerAccountController : Controller
    {

        AppDbContext _appdbcontext;
        public CustomerAccountController(AppDbContext ctxt)
        {
            _appdbcontext = ctxt;
        }

        [Authorize(Roles = "Admin")]
        [HttpPost("CreateCustomerAccount")]
        public IActionResult CreateCustomerAccount(int acc_num, int bal, DateOnly date, int cus_id)
        {
            CustomerAccount customerAccount = new CustomerAccount() { AccountNumber = acc_num, Balance = bal, CreatedDate = date, CustomerId = cus_id };
        }
    }
}
```

}
}
}

Details Controller

```
using Microsoft.AspNetCore.Authorization;
using Microsoft.AspNetCore.Mvc;
using MiniProject2.Context;
using MiniProject2.Models;
```

```
namespace MiniProject2.Controllers
```

```
{
public class CustomerDetailsController : Controller
{
    AppDbContext _appdbcontext;
    public CustomerDetailsController(AppDbContext ctxt)
    {
        _appdbcontext = ctxt;
    }
    [Authorize(Roles = "Admin")]
    [HttpPost("AddCustomerDetails")]
    public IActionResult AddCustomerDetails(string cus_name, int cus_age)
    {
        CustomerDetails customerDetail = new CustomerDetails() { CustomerName =
cus_name, CustomerAge = cus_age};
        _appdbcontext.customerDet.Add(customerDetail);
        _appdbcontext.SaveChanges();
        return Ok("Added successfully");
    }
    [HttpGet("GetCustomerDetails")]
    public IActionResult GetCustomerDetails()
    {
        var customer_details = from cd in _appdbcontext.customerDet
                                join ca in _appdbcontext.customerAcc
                                on cd.CustomerId equals ca.CustomerId
                                select new
                                {
                                    cd.CustomerId,
                                    cd.CustomerName,
                                    cd.CustomerAge,

```

```

        ca.CreatedDate,
        ca.AccountNumber,
        ca.Balance
    };
    return Ok(customer_details);
}
[Authorize(Roles = "Admin")]
[HttpPost("GetOverBalanceOfUser")]
public IActionResult GetOverBalanceOfUser(string cus_name)
{
    var cus_det = _appdbcontext.customerDet.FirstOrDefault(x => x.CustomerName ==
cus_name);

    var customer_balance = (from cd in _appdbcontext.customerDet
        join ca in _appdbcontext.customerAcc
        on cd.CustomerId equals ca.CustomerId
        where cd.CustomerName == cus_det.CustomerName
        select ca.Balance).Sum();
    return Ok(customer_balance);
}
}
}

```

User Controller

```

using Azure.Core;
using Microsoft.AspNetCore.Identity.Data;
using Microsoft.AspNetCore.Mvc;
using MiniProject2.Context;
using MiniProject2.Models;
using MiniProject2.Security;
using System.Data;
using System.IdentityModel.Tokens.Jwt;
using System.Security.Claims;

```

```

namespace MiniProject2.Controllers
{
    public class LoginRequest
    {
        public string userName { get; set; }
        public string password { get; set; }
    }
    public class UserLoginResponse
    {

```

```

        public string UserName { get; set; }
        public string UserRole { get; set; }
    }
    public class UserController : Controller
    {
        AppDbContext _appdbcontext;
        Microsoft.Extensions.Options.IOptions<JwtOptions> _jwtOptions;
        IHttpClientFactory _clientFactory;

        public UserController(AppDbContext ctxt,
        Microsoft.Extensions.Options.IOptions<JwtOptions> jwtOptions, IHttpClientFactory
        clientFactory)
        {
            _appdbcontext = ctxt;
            _jwtOptions = jwtOptions;
            _clientFactory = clientFactory;
        }

        [HttpPost("Login")]
        public async Task<ActionResult> Login(string _userName, string _password)
        {
            var client = _clientFactory.CreateClient();

            var request = new LoginRequest
            {
                userName = _userName,
                password = _password
            };
            var response = await client.PostAsJsonAsync("http://127.0.0.1:8000/login", request);

            if (response == null)
            {
                return Unauthorized("User Not Found");
            }
            else
            {
                var jsonResult = await response.Content.ReadAsStringAsync();

                var userData =
                System.Text.Json.JsonSerializer.Deserialize<UserLoginResponse>(jsonResult);

                if (userData == null)
                {

```

```

        return Unauthorized("Invalid login response");
    }

    var claims = new List<Claim>
    {
        new Claim(JwtRegisteredClaimNames.Name, userData.UserName)
    };

    if (!string.IsNullOrEmpty(userData.UserRole) &&
        userData.UserRole.Trim().ToLower().Contains("admin"))
    {
        claims.Add(new Claim(ClaimTypes.Role, "Admin"));
    }
    else
    {
        claims.Add(new Claim(ClaimTypes.Role, "User"));
    }

    var token = JwtService.CreateJWTToken(_jwtOptions.Value, claims);

    var customer_balance = (from cd in _appdbcontext.customerDet
                            join ca in _appdbcontext.customerAcc
                            on cd.CustomerId equals ca.CustomerId
                            where cd.CustomerName == userData.UserName
                            select ca.Balance).Sum();

    return Ok(new { token });
}
}
}
}

```

Account Model

```

using System.ComponentModel.DataAnnotations;
using System.ComponentModel.DataAnnotations.Schema;

namespace MiniProject2.Models
{
    public class CustomerAccount
    {
        [Key]
        public int AccountId { get; set; }
        public int AccountNumber { get; set; }
    }
}

```

```

        public int Balance { get; set; }
        public DateOnly CreatedDate { get; set; }
        public int CustomerId { get; set; }
        [ForeignKey("CustomerId")]
        public CustomerDetails customerDetails { get; set; }
    }
}

```

Customer Model

```
using System.ComponentModel.DataAnnotations;
```

```

namespace MiniProject2.Models
{
    public class CustomerDetails
    {
        [Key]
        public int CustomerId { get; set; }
        public string CustomerName { get; set; }
        public int CustomerAge { get; set; }
    }
}

```

Jwt Model

```

namespace MiniProject2.Models
{
    public class JwtOptions
    {
        public string Issuer { get; set; } = "";
        public string Audience { get; set; } = "";
        public string Key { get; set; } = "";
        public int ExpirationMinutes { get; set; } = 30;
    }
}

```

Program.cs

```

using Microsoft.AspNetCore.Authentication.JwtBearer;
using Microsoft.EntityFrameworkCore;
using Microsoft.IdentityModel.Tokens;
using MiniProject2.Models;
using System.Security.Claims;
using System.Text;

```

```
var builder = WebApplication.CreateBuilder(args);

// Add services to the container.

builder.Services.AddControllers();
// Learn more about configuring Swagger/OpenAPI at https://aka.ms/asptnetcore/swashbuckle
builder.Services.AddEndpointsApiExplorer();
builder.Services.AddSwaggerGen(c =>
{
    c.SwaggerDoc("v1", new() { Title = "SampleApi", Version = "v1" });

    // ?? Add JWT Bearer Authentication to Swagger
    c.AddSecurityDefinition("Bearer", new Microsoft.OpenApi.Models.OpenApiResponseScheme
        {
            Name = "Authorization",
            Type = Microsoft.OpenApi.Models.SecuritySchemeType.ApiKey,
            Scheme = "Bearer",
            BearerFormat = "JWT",
            In = Microsoft.OpenApi.Models.ParameterLocation.Header,
            Description = "Enter 'Bearer' [space] and then your valid token.\nExample: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXLTQyLWVudC9hdGVkLnNpdjkiLCJ0eXAiOiJKdW8iLCJpcyAibmFtZSI6ImNpbGljaSIsImxkaXI6ImNpbGljaSIsImVudCI6ImNpbGljaS..."}));

    c.AddSecurityRequirement(new Microsoft.OpenApi.Models.OpenApiResponseRequirement
        {
            {
                new Microsoft.OpenApi.Models.OpenApiResponseScheme
                    {
                        Reference = new Microsoft.OpenApi.Models.OpenApiResponseReference
                            {
                                Type = Microsoft.OpenApi.Models.ReferenceType.SecurityScheme,
                                Id = "Bearer"
                            }
                    },
                new string[] {}
            }
        });

});

var sql_section = builder.Configuration.GetSection("Sql_Connection");
string connectionString = sql_section["conn_str"];

builder.Services.Configure<JwtOptions>(builder.Configuration.GetSection("Jwt"));
var jwt = builder.Configuration.GetSection("Jwt");
```

```

var issuer = jwt["Issuer"];
var audience = jwt["Audience"];
var key = new SymmetricSecurityKey(Encoding.UTF8.GetBytes(jwt["key"]));

builder.Services.AddAuthentication(options =>
{
    options.DefaultAuthenticateScheme = JwtBearerDefaults.AuthenticationScheme;
    options.DefaultChallengeScheme = JwtBearerDefaults.AuthenticationScheme;

}).AddJwtBearer(options =>
{
    options.TokenValidationParameters = new TokenValidationParameters
    {
        ValidIssuer = issuer,
        ValidAudience = audience,
        IssuerSigningKey = key,
        ValidateIssuer = true,
        ValidateAudience = true,
        ValidateIssuerSigningKey = true,
        ValidateLifetime = true,
        ClockSkew = TimeSpan.Zero,
        RoleClaimType = ClaimTypes.Role,
        NameClaimType = ClaimTypes.NameIdentifier
    };

});

builder.Services.AddDbContext<MiniProject2.Context.AppDbContext>
(x => x.UseSqlServer(connectionString));
builder.Services.AddHttpClient();
var app = builder.Build();

// Configure the HTTP request pipeline.
if (app.Environment.IsDevelopment())
{
    app.UseSwagger();
    app.UseSwaggerUI();
}

app.UseHttpsRedirection();
app.UseAuthorization();
app.MapControllers();
app.Run();

```


Output

Curl

```
curl -X 'POST' \
'https://localhost:7114/Login?_userName=Dharanesh&_password=DharaneshX40123' \
-H 'accept: */*' \
-d ''
```

Request URL

```
https://localhost:7114/Login?_userName=Dharanesh&_password=DharaneshX40123
```

Server response

Code	Details
200	Response body<pre>{ "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ3YmV1IjoiaRGhhcmFuZDNoIiwiaHR0cDovL3NjaGVtYXNubWljcm9zb2Z0LmNvbS93cy8yMDA4LzA2L2lkZW50aXR5L2NsYWltcy9yb2x1Ijoia0NRtam41lCjleHA1OjE3NTU0NTgwRjQsImZcyI6Imh0dHBzOi8vd3d3LnBheW9kYS5jb20vIiwiaXVkiIjoia0VzZd9tZXJzX0NTaGFycCJ9.HMSkg0sNs2B0TINixT8CrIQXhf8AMwxJA_KqPM8s" }</pre> Response headers<pre>content-type: application/json; charset=utf-8 date: Fri, 22 Aug 2025 09:50:27 GMT server: Kestrel</pre>

Responses

Code	Description	Links
200	OK	No links

Curl

```
curl -X 'POST' \
'https://localhost:7114/GetOverBalanceOfUser?cus_name=Dharanesh' \
-H 'accept: */*' \
-d ''
```

Request URL

```
https://localhost:7114/GetOverBalanceOfUser?cus_name=Dharanesh
```

Server response

Code	Details
401	Error: response status is 401 Response headers<pre>content-length: 0 date: Fri, 22 Aug 2025 09:51:04 GMT server: Kestrel www-authenticate: Bearer</pre>

Responses

Code	Description	Links
200	OK	No links

Curl

```
curl -X 'POST' \
  'https://localhost:7114/GetOverBalanceOfUser?cus_name=Dharanesh' \
  -H 'accept: */*' \
  -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ3YW11Ijo1RGhhcmFuZlNoIiwiaHR0cDovL3NjaGVtYXMuYm91cm9zb2Z0LmVhbnV5S93cy8yMDA4LzA2L2lkZW50aXR5L2NsYWltcy9yb2x1Ijo1OQ' \
  -d ''
```

Request URL

https://localhost:7114/GetOverBalanceOfUser?cus_name=Dharanesh

Server response

Code	Details
------	---------

200

Response body

12000

Response headers

content-type: application/json; charset=utf-8
date: Fri, 22 Aug 2025 09:52:44 GMT
server: Kestrel

Responses

Code	Description
------	-------------

Links

Curl

```
curl -X 'GET' \
  'https://localhost:7114/GetCustomerDetails' \
  -H 'accept: */*' \
  -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ3YW11Ijo1RGhhcmFuZlNoIiwiaHR0cDovL3NjaGVtYXMuYm91cm9zb2Z0LmVhbnV5S93cy8yMDA4LzA2L2lkZW50aXR5L2NsYWltcy9yb2x1Ijo1OQ' \
  -d ''
```

Request URL

https://localhost:7114/GetCustomerDetails

Server response

Code	Details
------	---------

200

Response body

```
{
  "customerId": 1,
  "customerName": "dharanesh",
  "customerAge": 20,
  "createdDate": "0001-01-01",
  "accountNumber": 1002,
  "balance": 10000
},
{
  "customerId": 2,
  "customerName": "Karthi",
  "customerAge": 21,
  "createdDate": "0001-01-01",
  "accountNumber": 1003,
  "balance": 10000
}
```

Curl

```
curl -X 'POST' \
  'https://localhost:7114/AddCustomerDetails?cus_name=arun&cus_age=34' \
  -H 'accept: */*' \
  -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ3YW11Ijo1RGhhcmFuZlNoIiwiaHR0cDovL3NjaGVtYXMuYm91cm9zb2Z0LmVhbnV5S93cy8yMDA4LzA2L2lkZW50aXR5L2NsYWltcy9yb2x1Ijo1OQ' \
  -d ''
```

Request URL

https://localhost:7114/AddCustomerDetails?cus_name=arun&cus_age=34

Server response

Code	Details
------	---------

200

Response body

Added successfully

Response headers

content-type: text/plain; charset=utf-8
date: Fri, 22 Aug 2025 09:54:00 GMT
server: Kestrel

Curl

```
curl -X 'POST' \
  'https://localhost:7114/CreateCustomerAccount?acc_num=122&bal=20000&date=X222025-08-22X22&cus_id=4' \
  -H 'accept: */*' \
  -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1YVllIjoiRGhhcmFuZlZlbnR5IiwiaHR0cDovL3NjaGVtYXMuYm91cm9zb2Z0LmNvbS93cy8yMDA4LzA2L2lkZW50aXR5L2NsYWltcy9yb2x1IiwiaWF0Ijoi2025-08-22T09:55:03Z' \
  -d ''
```

Request URL

https://localhost:7114/CreateCustomerAccount?acc_num=122&bal=20000&date=X222025-08-22X22&cus_id=4

Server response

Code Details

200

Response body

Added successfully



Download

Response headers

content-type: text/plain; charset=utf-8
date: Fri, 22 Aug 2025 09:55:03 GMT
server: Kestrel

Responses